

ECON1050 Tutorial 8

Constrained Optimization

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Today's Plan

- ▶ Recap: key ideas from Lecture 8
- ▶ Tutorial questions: Q1–Q4

Recap: Constrained Optimization

General form

A constrained optimization problem has an objective function and an equality constraint:

$$\max (\min) f(x, y) \quad \text{subject to} \quad g(x, y) = c.$$

Key idea

Unlike unconstrained optimization, x and y cannot be chosen independently: feasible choices must satisfy the constraint.

Recap: The Lagrange Multiplier Method

Lagrangian

Combine the objective function and the constraint into

$$\mathcal{L}(x, y, \lambda) = f(x, y) - \lambda(g(x, y) - c).$$

Here λ is the Lagrange multiplier.

First-order conditions

A stationary solution satisfies

$$\mathcal{L}_x = f_x - \lambda g_x = 0, \quad \mathcal{L}_y = f_y - \lambda g_y = 0,$$

$$\mathcal{L}_\lambda = -(g(x, y) - c) = 0.$$

Solve these three equations simultaneously for x , y , and λ .

Recap: Interpreting the Lagrange Multiplier

Suppose the constraint is $g(x, y) = c$, and let $f^*(c)$ denote the optimal value of the objective function.

Shadow price

Under the regularity conditions used in the lecture,

$$\lambda(c) = \frac{df^*(c)}{dc}.$$

Therefore, λ measures how the optimal objective value changes when the resource in the constraint increases by one unit.

Economic interpretation

λ is the **shadow price** of the constrained resource.

Recap: Second-Order Conditions

For

$$\max (\min) f(x, y) \quad \text{s.t.} \quad g(x, y) = c,$$

let (x_0, y_0, λ) satisfy the FOCs and define

$$D = \mathcal{L}_{xx}(g_y)^2 - 2\mathcal{L}_{xy}g_xg_y + \mathcal{L}_{yy}(g_x)^2.$$

Local classification

$$D < 0 \implies \text{local maximum},$$

$$D > 0 \implies \text{local minimum}.$$

Recap: Concave and Convex Lagrangians

Concavity

If, throughout a convex set,

$$\mathcal{L}_{xx} \leq 0, \quad \mathcal{L}_{yy} \leq 0, \quad \mathcal{L}_{xx}\mathcal{L}_{yy} - \mathcal{L}_{xy}^2 \geq 0,$$

then the Lagrangian is concave. A stationary point solves the maximization problem globally.

Convexity

If

$$\mathcal{L}_{xx} \geq 0, \quad \mathcal{L}_{yy} \geq 0, \quad \mathcal{L}_{xx}\mathcal{L}_{yy} - \mathcal{L}_{xy}^2 \geq 0,$$

then the Lagrangian is convex. A stationary point solves the minimization problem globally.

Tutorial Q1

Question 1

Calculate the values of x_1 and x_2 that optimize the objective function subject to the constraint. Show that the solution is a local maximum.

$$\max y = 2x_1x_2 \quad \text{s.t.} \quad 2x_1 + x_2 = 10.$$

Question 1

Calculate the values of x_1 and x_2 that optimize the objective function subject to the constraint. Show that the solution is a local maximum.

$$\max y = 2x_1x_2 \quad \text{s.t.} \quad 2x_1 + x_2 = 10.$$

For

$$\max (\min) f(x, y) \quad \text{s.t.} \quad g(x, y) = c,$$

let (x_0, y_0, λ) satisfy the FOCs and define

$$D = \mathcal{L}_{xx}(g_y)^2 - 2\mathcal{L}_{xy}g_xg_y + \mathcal{L}_{yy}(g_x)^2.$$

Local classification

$$\boxed{D < 0 \Rightarrow \text{local maximum}}, \quad \boxed{D > 0 \Rightarrow \text{local minimum}}.$$

Tutorial Q1: Lagrangian and FOCs

Knowledge used

- ▶ Lagrange multiplier method for one equality constraint.
- ▶ First-order conditions in x_1 , x_2 , and λ .

Solution

$$\mathcal{L} = 2x_1x_2 - \lambda(2x_1 + x_2 - 10).$$

The FOCs are

$$2x_2 - 2\lambda = 0, \tag{1}$$

$$2x_1 - \lambda = 0, \tag{2}$$

$$2x_1 + x_2 - 10 = 0. \tag{3}$$

From (2), $\lambda = 2x_1$. Substituting into (1) gives $x_2 = 2x_1$. Using (3), $2x_1 + 2x_1 = 10$. Hence

$$\boxed{x_1^* = \frac{5}{2}, \quad x_2^* = 5, \quad \lambda = 5}.$$

Tutorial Q1: Second-Order Condition

Knowledge used

- Constrained second-order condition using D ; $D < 0$ confirms a local maximum.

Solution

For the Lagrangian and constraint,

$$\mathcal{L}_{11} = 0, \quad \mathcal{L}_{22} = 0, \quad \mathcal{L}_{12} = 2, \quad g_1 = 2, \quad g_2 = 1.$$

Therefore,

$$\begin{aligned} D &= \mathcal{L}_{11}(g_2)^2 - 2\mathcal{L}_{12}g_1g_2 + \mathcal{L}_{22}(g_1)^2 \\ &= 0 - 2(2)(2)(1) + 0 = -8 < 0. \end{aligned}$$

Thus,

$\left(\frac{5}{2}, 5\right)$ is a local maximum.

Question 2

Calculate the values of x_1 and x_2 that optimize the objective function subject to the constraint. Show that the solution is a local minimum, and then show that it is a global minimum.

$$\min y = x_1^2 + 2x_2^2 - 2x_1x_2 \quad \text{s.t.} \quad 2x_1 - 7x_2 + 29 = 0.$$

Question 2

Calculate the values of x_1 and x_2 that optimize the objective function subject to the constraint. Show that the solution is a local minimum, and then show that it is a global minimum.

$$\min y = x_1^2 + 2x_2^2 - 2x_1x_2 \quad \text{s.t.} \quad 2x_1 - 7x_2 + 29 = 0.$$

Concavity and convexity

If, throughout a convex set,

$$\mathcal{L}_{xx} \leq 0, \quad \mathcal{L}_{yy} \leq 0, \quad \mathcal{L}_{xx}\mathcal{L}_{yy} - \mathcal{L}_{xy}^2 \geq 0,$$

then the Lagrangian is concave. A stationary point solves the maximization problem globally.

If

$$\mathcal{L}_{xx} \geq 0, \quad \mathcal{L}_{yy} \geq 0, \quad \mathcal{L}_{xx}\mathcal{L}_{yy} - \mathcal{L}_{xy}^2 \geq 0,$$

then the Lagrangian is convex. A stationary point solves the minimization problem globally.

Tutorial Q2: Lagrangian and FOCs

Knowledge used

- ▶ Lagrangian with a linear equality constraint.
- ▶ Solve the three FOCs simultaneously.

Solution

$$\mathcal{L} = x_1^2 + 2x_2^2 - 2x_1x_2 - \lambda(2x_1 - 7x_2 + 29).$$

The FOCs are

$$2x_1 - 2x_2 - 2\lambda = 0, \tag{1}$$

$$4x_2 - 2x_1 + 7\lambda = 0, \tag{2}$$

$$2x_1 - 7x_2 + 29 = 0. \tag{3}$$

From (3), $x_1 = \frac{7}{2}x_2 - \frac{29}{2}$. Substituting into (1) and (2) gives

$$\boxed{x_1^* = 3, \quad x_2^* = 5, \quad \lambda = -2}.$$

Tutorial Q2: Local Minimum

Knowledge used

- ▶ Apply the constrained second-order condition.
- ▶ $D > 0$ confirms a local minimum.

Solution

The relevant derivatives are

$$\begin{aligned}\mathcal{L}_{11} &= 2, & \mathcal{L}_{22} &= 4, & \mathcal{L}_{12} &= -2, \\ g_1 &= 2, & g_2 &= -7.\end{aligned}$$

Hence

$$\begin{aligned}D &= 2(-7)^2 - 2(-2)(2)(-7) + 4(2)^2 \\ &= 98 - 56 + 16 \\ &= 58 > 0.\end{aligned}$$

Therefore,

$(x_1^*, x_2^*) = (3, 5)$ is a local minimum.

Tutorial Q2: Global Minimum

Knowledge used

- Convexity of the Lagrangian provides a sufficient condition for a global minimum.

Solution

For all (x_1, x_2) ,

$$\mathcal{L}_{11} = 2 > 0, \quad \mathcal{L}_{22} = 4 > 0,$$

and

$$\mathcal{L}_{11}\mathcal{L}_{22} - \mathcal{L}_{12}^2 = (2)(4) - (-2)^2 = 4 > 0.$$

Thus the Lagrangian is convex. Therefore the stationary solution is also the global minimum:

$(x_1^*, x_2^*) = (3, 5) \text{ is the global minimum.}$

Question 3

A competitive firm produces two products, q_1 and q_2 , with fixed prices

$$p_1 = 15, \quad p_2 = 12.$$

Its joint total cost is

$$C = 20 - 4q_1q_2 + 7q_1 + 6q_2,$$

and production is constrained by

$$2q_1 + 4q_2 = 100.$$

Find the profit-maximizing outputs and maximum profit, confirm the maximum, and interpret λ .

Tutorial Q3: Profit Function and Lagrangian

Knowledge used

- ▶ Profit is $\pi = TR - TC$.
- ▶ Apply constrained optimization to the profit function.

Solution

Total revenue is

$$TR = 15q_1 + 12q_2.$$

Therefore,

$$\begin{aligned}\pi(q_1, q_2) &= (15q_1 + 12q_2) - (20 - 4q_1q_2 + 7q_1 + 6q_2) \\ &= 8q_1 + 6q_2 + 4q_1q_2 - 20.\end{aligned}$$

The Lagrangian is

$$\mathcal{L} = 8q_1 + 6q_2 + 4q_1q_2 - 20 - \lambda(2q_1 + 4q_2 - 100).$$

Tutorial Q3: First-Order Conditions

Knowledge used

- Set $\mathcal{L}_{q_1} = \mathcal{L}_{q_2} = \mathcal{L}_\lambda = 0$ and solve simultaneously.

Solution

The FOCs are

$$8 + 4q_2 - 2\lambda = 0, \quad (1)$$

$$6 + 4q_1 - 4\lambda = 0, \quad (2)$$

$$2q_1 + 4q_2 - 100 = 0. \quad (3)$$

From (1), $\lambda = 4 + 2q_2$. Substituting into (2) gives $q_1 = \frac{5}{2} + 2q_2$. Using the constraint then gives $8q_2 - 95 = 0$. Hence

$$\boxed{q_2^* = \frac{95}{8}, \quad q_1^* = \frac{105}{4}}.$$

Tutorial Q3: Maximum Profit

Knowledge used

- Substitute the optimal outputs into the profit function.

Solution

At

$$q_1^* = \frac{105}{4}, \quad q_2^* = \frac{95}{8},$$

maximum profit is

$$\begin{aligned} \pi_{\max} &= 8 \left(\frac{105}{4} \right) + 6 \left(\frac{95}{8} \right) - 20 + 4 \left(\frac{105}{4} \right) \left(\frac{95}{8} \right) \\ &= \boxed{\frac{12065}{8}} \approx 1508.125. \end{aligned}$$

Tutorial Q3: Confirm the Maximum

Knowledge used

- Use the constrained second-order condition to classify the stationary point.

Solution

Here

$$\mathcal{L}_{11} = 0, \quad \mathcal{L}_{22} = 0, \quad \mathcal{L}_{12} = 4,$$

and

$$g_1 = 2, \quad g_2 = 4.$$

Therefore,

$$\begin{aligned} D &= 0(4)^2 - 2(4)(2)(4) + 0(2)^2 \\ &= -64 < 0. \end{aligned}$$

Thus,

$\left(\frac{105}{4}, \frac{95}{8}\right)$ is a local maximum.

Tutorial Q3: Interpret λ

Knowledge used

- ▶ λ is the shadow price of the constrained resource.
- ▶ Here the constrained resource is production time.

Solution

From the FOC,

$$\lambda = 4 + 2q_2.$$

At the optimum,

$$\lambda^* = 4 + 2\left(\frac{95}{8}\right) = \frac{111}{4} = 27.75.$$

Hence,

$$\boxed{\lambda^* = 27.75}.$$

Interpretation: if one additional hour of production time becomes available, the maximized profit increases by approximately 27.75 units of profit.

Question 4

A consumer has utility

$$U(x_1, x_2) = 2 \ln(x_1) + \ln(x_2 - 2)$$

and budget constraint

$$y = p_1 x_1 + p_2 x_2.$$

Derive the demand functions for the two goods. Then, for

$$p_1 = 4, \quad p_2 = 2, \quad y = 100,$$

calculate the quantities purchased.

Tutorial Q4: Lagrangian and FOCs

Knowledge used

- ▶ Utility maximization subject to a budget constraint.
- ▶ Lagrange multiplier method and logarithmic derivatives.

Solution

The Lagrangian is

$$\mathcal{L} = 2 \ln(x_1) + \ln(x_2 - 2) - \lambda(p_1 x_1 + p_2 x_2 - y).$$

The FOCs are

$$\frac{2}{x_1} - \lambda p_1 = 0, \tag{1}$$

$$\frac{1}{x_2 - 2} - \lambda p_2 = 0, \tag{2}$$

$$p_1 x_1 + p_2 x_2 - y = 0. \tag{3}$$

From (1),

$$\lambda = \frac{2}{p_1 x_1}.$$

Tutorial Q4: Derive x_2^*

Knowledge used

- ▶ Use the budget constraint to eliminate one variable.
- ▶ Combine the two marginal conditions.

Solution

From the budget constraint,

$$x_1 = \frac{y - p_2 x_2}{p_1}.$$

Combining FOCs (1) and (2),

$$\frac{1}{x_2 - 2} = \frac{2p_2}{p_1 x_1}.$$

Substituting for x_1 gives

$$y - p_2 x_2 = 2p_2 (x_2 - 2) \implies y + 4p_2 = 3p_2 x_2.$$

Hence

$$x_2^* = \frac{y + 4p_2}{3p_2}.$$

Tutorial Q4: Demand Functions

Knowledge used

- ▶ Substitute the optimal x_2 back into the budget constraint.
- ▶ Express optimal choices as functions of income and prices.

Solution

Substitute x_2^* into the budget constraint:

$$\begin{aligned}x_1^* &= \frac{y - p_2 \left(\frac{y + 4p_2}{3p_2} \right)}{p_1} \\&= \frac{2(y - 2p_2)}{3p_1}.\end{aligned}$$

Therefore, the demand functions are

$$\boxed{x_1^* = \frac{2(y - 2p_2)}{3p_1}, \quad x_2^* = \frac{y + 4p_2}{3p_2}}.$$

Tutorial Q4: Confirm the Maximum

Knowledge used

- ▶ Use the constrained second-order condition.
- ▶ Negative D implies a local maximum.

Solution

The second derivatives are

$$\mathcal{L}_{11} = -\frac{2}{x_1^2}, \quad \mathcal{L}_{22} = -\frac{1}{(x_2 - 2)^2}, \quad \mathcal{L}_{12} = 0,$$

and

$$g_1 = p_1, \quad g_2 = p_2.$$

Therefore,

$$\begin{aligned} D &= -\frac{2}{x_1^2} p_2^2 - \frac{1}{(x_2 - 2)^2} p_1^2 \\ &= -2 \left(\frac{p_2}{x_1} \right)^2 - \left(\frac{p_1}{x_2 - 2} \right)^2 < 0. \end{aligned}$$

Hence the stationary solution is a local maximum.

Tutorial Q4: Numerical Solution

Knowledge used

- Substitute the given prices and budget into the demand functions.

Solution

For

$$p_1 = 4, \quad p_2 = 2, \quad y = 100,$$

$$x_1^* = \frac{2(100 - 2 \cdot 2)}{3 \cdot 4} = \frac{192}{12} = 16,$$

$$x_2^* = \frac{100 + 4 \cdot 2}{3 \cdot 2} = \frac{108}{6} = 18.$$

Therefore,

$$\boxed{x_1^* = 16, \quad x_2^* = 18}.$$

Key Takeaways

- ▶ Equality-constrained optimization can be solved by forming a Lagrangian and applying the FOCs.
- ▶ The Lagrange multiplier λ is the shadow price of the constrained resource.
- ▶ The constrained second-order expression D distinguishes local maxima from local minima.
- ▶ Concavity or convexity of the Lagrangian can establish a global optimum.
- ▶ Economic applications include profit maximization and utility maximization subject to resource constraints.